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The Impacts of Offshore Wind Farms on Coastal Tourists' Behaviors in South Korea

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ABSTRACT
Offshore wind power has grown significantly worldwide but the presence of wind farms can be problematic in tourism areas. Offshore wind farms affect tourism behaviors. This study was conducted to assess the extent to which individuals' characteristics affect their acceptance of wind farm projects, tourism behaviors represented as trip intention and site substitution when the offshore wind turbines are deployed in the beach. Using a case of offshore wind farm projects in South Korea, study results show that individuals' perceptions of wind energy and perceived impacts of tourism were the critical variables that help understand the public acceptance. The likelihood of an individual visiting a beach with offshore wind farms was connected to their values concerning wind energy, seascape, and the environment. Such values played an important role in deciding to visit a different site. The results expectedly provide meaningful implications related to the siting of offshore wind farms.

KEYWORDS
offshore wind power; public acceptance; coastal tourism; tourism impacts; tourism behaviors

Introduction

A search for clean renewable energy sources has become almost mandatory for long term protection of the environment. In contrast to conventional energy sources, renewable energy sources such as wind power, nuclear power, and hydropower are often considered benign to the environment. Particularly, the utilization of wind energy and offshore wind power has grown significantly worldwide over the last decades as a result of supportive policies and a significant decrease in the cost of constructing the necessary infrastructure (Park and Kim 2019). In line with this trend, the South Korean government developed the Renewable Energy 2030 Plan that aims to expand the share of renewable energy from 2.2% in 2016 to 20% in 2030 (Kim, Nam, and Yoo 2020). The popularity of offshore wind farms has increased and projects are currently ongoing in 67 locations (Korea Wind Energy Industry Association 2021). However, there are also significant concerns regarding the negative impacts associated with the development of offshore wind farms. These concerns include, but not limited to, the visual

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disruption of landscapes, erosion, the possible endangerment of wildlife ecosystems, and adverse effects on coastal tourism (Goodale and Milman 2019; Ladenburg 2009; Rudolph 2014). These potential problems have made the development of offshore wind farm projects a subject of contested debates on local, regional and national levels in many countries including Korea.

The development of offshore wind farms can become more problematic in areas where tourism is prosperous and various stakeholders are involved. Firestone and Kempton (2007) showed that the introduction of wind farms into tourism areas can be a threat to coastal qualities encompassing the esthetic of the ocean view, recreational boating, community harmony, and other marine environments, which may lead to damaging tourism related businesses. Because tourism relies heavily on visual images of the place (Urry 1990), the presence of offshore wind turbines may spoil tourists' needs for unobstructed landscape (Rudolph 2014). However, the negative impacts have not been validated through empirical research. On the contrary, several studies (Westerberg, Jacobsen, and Lifran 2013; Westerberg, Bredahl, and Lifran 2015) provided results that the presence of offshore wind farms is not a deterrent but an attractant for tourists. Rudolph (2014) contends that the negative impacts of offshore wind farms on tourism are socially constructed particularly through five basic themes: visual disruptions, disruptions of local identities, a certain image of tourists, disturbance of recreational activities, and negative environmental impacts. A socially constructed nature regarding the impacts of offshore wind farms on coastal tourism appears to make potential tourists' acceptance of offshore wind farm projects and subsequent coping behaviors (i.e., site substitution and trip intention) a crucial issue (Ferguson et al. 2021). Identifying the determinants of tourists' acceptance of offshore wind farms and their tourism related coping behavioral intentions is a pivotal step to advance our understanding of the controversial relationship between the development of offshore wind farms and coastal tourism.

A basic explanation for people's acceptance of offshore wind farms and/or tourism intentions on the site of development is that they compare perceived personal and/or societal costs and benefits from the development. However, such a dichotomous approach does not adequately explain what factors affect people's perceptions toward the costs and benefits. Thus, it calls for an effort to examine the factors that underlie the attitudes (Bidwell 2013). Several attitudinal factors that affect tourists' acceptance of offshore wind farms and their tourism related behavioral intentions include people's attitudes toward wind power (Wustenhagen, Wolsink, and Burer 2007), environmental attitudes (Bidwell 2017), and perceived visual impacts of the development on seascape (Bishop and Miller 2007; Krueger, Parsons, and Firestone 2011). Although these variables are potentially important determinants for people's support for the development, some other ones could be also important to reflect a tourism context. Therefore, people's general attitudes toward tourism impacts and anticipated effects of offshore wind farms on local tourism also need to be incorporated into the study.

This study was conducted to investigate the acceptance of offshore wind farm projects, particularly in Korea, as well as the intention of potential and current tourists to visit beaches where offshore wind farms are located. Specifically, this study attempts to assess the extent to which tourists' perceptions affect acceptance of offshore wind

farm projects and tourism behaviors such as the intention to visit a particular site (i.e., trip intention) and the intention to choose another site when offshore wind turbines are deployed on a beach that they have visited previously (i.e., site substitution). Tourists' perceptions utilized in this study include attitudes toward wind power, environmental attitudes, attitudes toward tourism impacts, impacts of offshore wind farms on local tourism and seascape. The results from the study are expected to contribute to a better understanding of tourists as important stakeholders in the development of wind farm projects, aiding policymakers in coping with various challenges regarding the adoption of clean energy systems while maintaining the long-term vitality of a region as a competitive tourist destination.

Literature review

Offshore wind farms and public acceptance

Public acceptance is a part of the broader concept of social acceptance, which is further divided into three dimensions: socio-political acceptance, community acceptance, and market acceptance (see Wustenhagen, Wolsink, and Burer 2007 for a detailed discussion). Particularly, public acceptance is on the socio-political dimension of social acceptance. Public acceptance within the context of offshore wind farms refers to the attitudinal responses of stakeholders who are influenced by the installment of offshore wind farms. This study focuses on potential and current tourists as key stakeholders of offshore wind farms. Although the offshore wind energy sector tends to receive strong support from central and local governments, gaining support from the public including tourists is considered a complicated task because of the many pros and cons surrounding the installment of wind turbines. In particular, local opposition to the development of wind farms has been explained through the concept "not in my backyard" (NIMBY), which represents an attitude-behavior gap on social and individual levels (Bell, Gray, and Haggett 2005). However, the NIMBY approach has been increasingly rejected for its low explanatory value (Rudolph 2014). Identifying the factors that influence objection or support (e.g., Ellis, Barry, and Robinson 2007; Devine-Wright and Howes 2010; Upham 2009) can be a better alternative to the simplistic NIMBY approach.

Offshore wind farms and tourism

There is a close connection between the development of offshore wind farms and tourism because the deployment of wind turbines is believed to greatly affect the visual perception of landscapes in which tourism resorts are located (Liu, Upchurch, and Curtis 2016). Despite the popular belief that the installment of wind turbines diminishes the view of a landscape and negatively affects the preferences of tourists, previous studies have produced mixed results regarding the impacts of wind turbines on tourism. Some studies have reported a negative change in tourism behaviors such as not visiting an area after a wind farm has been constructed because they believe the area will fail to provide the experience they are seeking (Broekel and Alfken 2015; Devine-Wright 2009). A study by Teisl et al. (2018) also hinted at the negative impacts of wind farms.

When visual information is provided via a virtual reality tool, people tend to display more negative reactions to the installment of offshore wind turbines. This suggests that the changes to the seascape that result from the presence of wind turbines can negatively affect the perceptions and behaviors of potential tourists. Dalton et al. (2020) used a stated preference approach to show that recreational boaters evaluate their experiences in areas with offshore wind farms as negative particularly when the photos of wind turbines too close to the sites were presented.

Evidences, however, supporting the actual negative consequences of offshore wind farms on tourism are relatively scarce and even contradicting results are reported (Rudolph 2014). For instance, Ólafsdóttir and Saeþórsdóttir (2019) studied offshore wind farm projects in Southern Iceland and suggest that wind turbines that have transformed the appearance of a wilderness area can attract different types of tourists with different needs and values. Carr-Harris and Lang (2019) estimated the impacts of offshore wind farms on tourism using AirBnB data to investigate rental demand and found that the presence of a wind farm caused a significant increase in nightly reservations, occupancy rates, and monthly revenue during the peak season. Smythe, Bidwell, and Tyler (2021) examined the impacts of offshore wind farms on the experiences of recreational anglers and found that anglers tend to favorably view wind farms as a symbol of green energy and claim that wind farms do not conflict with fishing, which supports the view that the installment of offshore wind farms does not necessarily conflict with tourism. Voltaire and Koutchade (2020) investigated the impacts of offshore wind turbines on the demand for beach tourism in Catalonia and showed that the deployment of wind turbines decreases the demand for beach tourism, while other beaches in the same region become more popular. This emphasizes the substitution effects that wind turbines may have on tourists. Westerberg, Bredahl, and Lifran (2015) argued that visits to an area with an offshore wind farm depend not only on the effects that wind turbines have on the view but on other factors such as the values of an individual in terms of climate change and attitudes toward wind energy. These contradictory results suggest that more research efforts are needed to better understand people's perceptions and behaviors regarding the offshore wind farms particularly within the context of tourism.

Determinants affecting tourism behaviors in relation to offshore wind farms

Understanding critical determinants shaping individuals' acceptance of offshore wind farms and subsequent tourism behaviors can be fundamental in devising effective developments and marketing plans regarding the development of offshore wind farms in a tourism area.

Perceptions of wind power and seascape

Individuals' attitudes toward wind turbines seem to depend on the perceived weight of the costs against the benefits (Smythe et al. 2020). Although it is evident that wind turbines change the physical landscape, the perceived impacts caused by the development of offshore wind farms can either reduce or enhance this perception depending on the manner by which the costs and benefits are assessed. Particularly, offshore

wind turbines can be considered attractive when perceived as a novel sight by tourists (Westerberg, Jacobsen, and Lifran 2013).

Despite the longstanding belief that offshore wind farms diminish the visual quality of the landscape, empirical results gathered over the past few decades have generated rather mixed results, particularly in terms of perception and behavior. Some of the negative impacts reported in previous studies include the refusal to visit an area following wind farm development because of the view (Broekel and Alfken 2015; Devine-Wright 2009) leading to a negative evaluation of the landscape and changes in the preference for an area (Teisl et al. 2018). However, contradictory results have also been reported (Carr-Harris and Lang 2019). Although it is not yet decided whether the deployment of offshore wind farms is particularly good or bad for the view of a coastal area, it is relatively clear that offshore wind farms can be pivotal in affecting the way people perceive a seascape. A positive view of seascapes with wind turbines will lead to positive reactions (e.g., acceptance of wind farms, increased intention to visit the area) and vice versa.

Environmental attitudes

Environmental attitudes refer to people's general thoughts about the relationship between humans and nature. The new environmental paradigm (NEP) may be the most popular construct with which to assess the environmental orientation of an individual (Dunlap and Van Liere 1978). The NEP scale focuses on the biospheric and anthropocentric perspectives of people toward nature. Individuals with more biospheric orientation are considered pro-environmental whereas those with a more anthropocentric orientation accept human domination over nature and support the development of nature for human needs. The NEP has been a useful tool to evaluate environmental attitudes and behaviors (e.g., Kim, Borges, and Chon 2006). A positive relationship between pro-environmental orientation and environmentally conscious behavior (e.g., recycling, petitioning, and energy conservation) has repeatedly been confirmed in previous empirical research (e.g., Stern et al. 1999). According the value-belief-norm theory, which posits a hierarchical relationship among cognitive structures, high-order beliefs (or attitudes) about environment such as NEP can affect related attitudes and behaviors because those cognitive structures are in a hierarchical relationship (Bidwell 2017). Viewed from the seemingly apparent positive relationship between environmental attitudes and behavior, it is deemed appropriate to suggest a positive causal relationship between pro-environmental attitudes and the acceptance of offshore wind farms because wind energy is thought to be environmentally benign and the installment of offshore wind turbines can contribute to the mitigation of global warming.

Results from the previous studies indicate a close relationship between environmental attitudes and the acceptance of wind farm development (Devine-Wright 2009; Fergen and Jacquet 2016). Individuals with a biospheric (pro-environmental) attitude are more likely to accept local wind farm developments, not only to protect the environment, but also because wind farms may produce local economic benefits and create novel esthetic value (Ellis, Barry, and Robinson 2007). Although previous studies have examined the effects that the environmental attitudes of an individual has on his/her tourism behavior (Imran, Alam, and Beaumont 2014; Kim, Borges, and Chon 2006), few have

investigated the relationship within the context of wind farm development. Given that tourism may be considered an anthropocentric activity, it could be inferred that individuals with an anthropocentric attitude may display tourism friendly intention and behaviors, while those with a biospheric attitude are less interested in visiting a tourism destination with attractions and developments nearby. However, the dearth of empirical research in relation to this issue makes it hard to present any concrete prediction. An empirical examination is needed to discover the causal relationship, particularly within the context of offshore wind farm development.

Attitudes toward tourism impacts

Perceived impacts of tourism on the local community can act as a significant stimulant to individuals' tourism related decisions and behaviors including visit intention to the area where offshore wind farms are located. Similar to environmental attitudes, individuals tend to have general perceptions toward tourism (Josiassen, Kock, and Norfelt 2022). The attitudes that people have about tourism in general can be a significant determinant in accepting offshore wind farms and their intention to visit such areas. Given that tourism as a social phenomenon encompasses the economic, environmental, and social dimensions, individuals can have different attitudes toward tourism from the perspective of several differing aspects (Ap and Crompton 1998). This study focuses mainly on economic and social dimensions of tourism impacts. Individuals who perceive greater economic benefits from tourism tend to have favorable attitudes toward tourism and tourism-related developments (Wang and Pfister 2008).

Thus, it is reasonable to assume that positive perceptions derived from the economic benefits of tourism can positively impact the tourism-related attitudes and behaviors surrounding offshore wind farms. Similar logic can be applied to the social dimension of tourism impacts. A social dimension of tourism impacts is generally related to how tourism affects social values of destination community. Those with favorable attitudes toward tourism from the social perspective may be more proactive in participating in tourism. Regarding offshore wind farms, people who perceive tourism as a positive catalyst for the social prosperity of a region will generally be more supportive of visiting coastal areas in which offshore wind farms are located. In a similar vein, if tourists perceive the offshore wind farms positively affects local tourism, they tend to support for the development of offshore wind farms and visit the sites of development. This is because a support for offshore wind farms largely depends on the belief that the projects provide economic benefits to the community (Bidwell 2013) and tourism may be perceived as a catalyst for economic prosperity.

Methods

Sampling

The data were collected through an online survey using a panel of Korean citizens recruited by a survey company in June 2018. A quota sampling approach was used in accordance with the population of each metropolitan region (a total of 17 metropolitan regions in Korea) to select a nationally representative sample. The study

population was individuals who had visited a beach at least once during the previous 12 months, as the questionnaire comprised offshore wind farm project-related questions based on their coastal trip experiences. A total of 82 million people visited the beaches of Korea in 2018 (Statistics of Korea 2020) and thus we assume that most of Koreans visited beaches at least once during this time period. A total of 1,695 invitations were sent out. Among them, 568 were not qualified because of the age, gender and regional quotas and another 71 did not complete the survey. The survey was completed after 1,056 individuals had completed the questionnaire.

Data analyses

Different regression models were employed to determine which explanatory variables are important in explaining limited dependent variables such as individuals' acceptance of offshore wind farm projects and their trip intention. For the dependent variables, the three different questions of public acceptance, site substitution and trip intention were employed from previous literature (e.g., Kim, Nam, and Yoo 2020 for public acceptance, Sutton and Ditton 2005 for site substitution and Oh and Hammitt 2011 for trip intention). First, to measure the public acceptance of offshore wind farm projects, respondents were asked "*would you support the project of the installation of offshore wind farms?*" to select the level of agreement using a 5-point rating scale ranging from (1) Strongly disagree to (5) Strongly agree. The respondents were then asked "*if the offshore wind farms are installed near the beach you have visited frequently, how would you change the number of trips to the beach?*", to which answered this question using a 3-point rating scale ranging from (1) decrease the number of trips to (3) increase the number of trips. Finally, with respect to site substitution, the respondents were asked "*If offshore wind farms are installed near the beach you have visited frequently, what would you do? Please read all five options below and then choose the one that best describes what you would do.*" This question was designed to measure individuals' site substitution as a trip intention (i.e., willingness to substitute) as a proxy of actual behavior. The five different options included: (1) I would come back to the same beach destination regardless of the installation of wind turbines; (2) I would go to another nearby beach destination in the same city or county; (3) I would go to another distant beach destination in a different city or county; (4) I would go to another destination that does not have a beach; and (5) I would not take trips to any destinations.

Non-linear regression models are usually employed when the dependent variable is a discrete outcome. The three dependent variables used in this study contained an ordinal variable for public acceptance of offshore wind farms, a binary variable for trip intention and a nominal variable for site substitution. Three different regression models are suitable for data analysis with limited dependent variables: the ordered probit model for the public acceptance of wind farms; the logit model for the intention to visit an area; and the multinomial logit model for site substitution.

Explanatory variables

Various independent variables were included in the limited dependent variable models, although not all were used in each model. As shown in Figure 1, these variables included

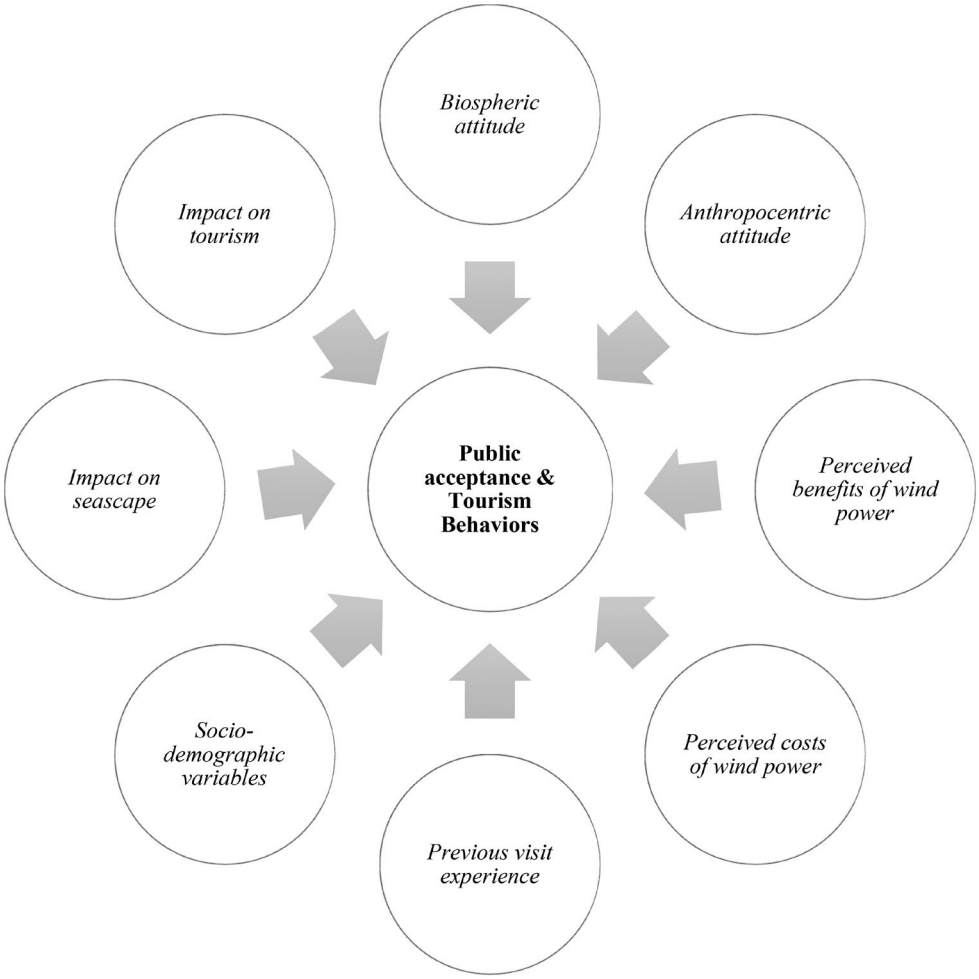

Figure 1. The proposed research models for individuals' acceptance of offshore wind farms and tourism behaviors. Note: dependent variables in bold and explanatory variables in italics. Tourism behaviors include trip intention and site substitution

individuals' perceptions of wind energy, environmental attitudes, attitudes toward tourism and impacts of offshore wind farms on local tourism and seascape as well as socio-demographic variables. Individuals' perceptions of wind energy were measured using ten items adapted from Koetin and Min (2014) study, using a 5-point rating scale that ranged from (1) Strongly disagree to (5) Strongly agree. To construct the variables, exploratory factor analysis (EFA) with principal component analysis was conducted and two factors, *perceived benefits* and the *perceived costs*, were identified. Seven items (e.g., "Wind power contributes to local economic development) were used for perceived benefits and three items were used for perceived costs (e.g., "Wind power destroys animals' and plants' habitats"). The results of the EFA showed that scale reliabilities were high for perceived benefits (Cronbach' α =.841) and perceived costs (α=.774).

Environmental attitudes were measured with 12 items derived from Dunlap and Van Liere's new environmental paradigm scale (Dunlap and Van Liere 1978). The scale

was developed to measure the three factors of *balance of nature* (e.g., "Humans are severely abusing the environment"), *limits to growth* (e.g., There are limits to growth beyond which our industrialized society cannot expand), and *human domination* (e.g., "Humans have the right to modify the natural environment to suit their needs") using a 5-point rating scale ranging from (1) Strongly disagree to (5) Strongly agree. The EFA supported the two factors, balance of nature (biospheric attitude) and human domination (anthropocentric attitude) after limits to growth was combined into the latter factor. Cronbach's alpha coefficients were 0.831 for balance of nature and 0.771 for human domination.

Attitudes toward tourism were measured with 24 items using Lankford and Howard (1994) tourism impact attitude scale. The scale was developed to measure positive and negative economic, social and environmental impacts of tourism on the local community but two factors which were considered to be closely related to the installation of offshore wind farms, were included for data analysis. The two factors of *economic impacts of tourism* (e.g., "Tourism increases household income") and *social impacts of tourism* (e.g., "Tourism increases the convenience of public transportation") were measured using a 5-point rating scale ranging from (1) Strongly disagree to (5) Strongly agree. The EFA supported these two factors. Cronbach's alpha coefficients were 0.869 for economic benefits and 0.828 for social benefits. The variable of each factor was created by summing up individual items for further data analysis.

We also included individuals' evaluations of the impacts of offshore wind farms on local tourism and seascape as well as socio-demographic variables to assess the importance of these variables in the models. Impacts of offshore wind farms on local tourism and seascape were measured with a single question using a 4-point rating scale ranging from (1) negative impact to (3) positive impact besides (4) not sure. Age was measured as a continuous variable and income was measured with 9 ordinal categories. In the data analysis, *income* was recoded as 1 if income was 6 million Korean Won per month (KRW; $5,378 as of 2018, 1115.7 KRW = $1 based on the statistics of the Bank of Korea) or more and 0 if income was less than 6 million KRW. *Gender* was specified as 1 for male and 0 for female. Respondents' residence in the capital region, *Capital*, was included as a dummy variable in which 1 is the capital region and 0 otherwise. Finally, respondents' average monthly electricity *bill* was measured with 12 ordinal categories (1 = 10,000 KRW - $9.0 and less, 12 = 200,000 KRW - $179.2 and above). Table 1 shows the definition of each variable included in the analysis.

All the data analysis was conducted using SPSS 25.0 and STATA 15.1 for the three different regression models with limited dependent variables. To explain the public acceptance of an offshore wind farm project, the ordered probit model was estimated with the following 13 explanatory variables: perceptions of wind power (perceived benefits, perceived costs), environmental attitudes (balance of nature, human domination), attitudes toward tourism (economic impacts of tourism, social impacts of tourism), age, gender, marital status, income, capital, visit and bill. The logit model and the multinomial logit model were estimated to examine individuals' trip intention to visit the beach and to examine site substation with the following explanatory variables: perceived benefits, perceived costs, balance of nature, human domination, impacts on tourism and impacts on seascape as well as socio-demographic variables.

Table 1. Explanatory variables used in the data analyses.

Variable names	Descriptions	Mean	%[a]	Standard deviation[b]
Perceived benefits	Positive perceptions of wind power (Coded 1 to 5: 1= strongly disagree, 5 = strongly agree)	3.749		0.53
Perceived costs	Negative perceptions of wind power (Coded 1 to 5: 1= strongly disagree, 5 = strongly agree)	3.158		0.75
Balance of nature	Biospheric attitude (Coded 1 to 5: 1= strongly disagree, 5 = strongly agree)	4.068		0.50
Human domination	Anthropocentric attitude (Coded 1 to 5: 1= strongly disagree, 5 = strongly agree)	2.801		0.59
Economic impacts of tourism	Economic impacts of tourism (Coded 1 to 5: 1= strongly disagree, 5 = strongly agree)	3.233		0.68
Social impacts of tourism	Social impacts of tourism (Coded 1 to 5: 1= not at all important, 5 = extremely important)	3.515		0.62
Impacts on tourism	Impacts of offshore wind farms on local tourism (Coded 1 to 5: 1= extremely negative, 5 = extremely positive)	3.737		0.72
Impacts on seascape	Impacts of offshore wind farms on seascape (Coded 1 to 5: 1= extremely negative, 5 = extremely positive)	2.529		0.60
Age	Age	42.847		12.25
Gender	Gender (1 = male, 0 = female)		51.2	
Married	Marital status (1 = married, 0 = not married)		67.4	
Income	Household income/month (1 = if income is 6 million KRW and more, 0 = if income is less than 6 million)		23.5	
Capital	Residence (1 = live in the capital region, 0 = otherwise)		50.8	
Visit	Previous experience of wind farms (1 = yes, 0 = no)		52.5	
Bill	Monthly household electricity bill (1= if bill is 50,000 KRW and more, 0 = less than 50,000 KRW)		30.9	

[a]percentage for dichotomous variables
[b]standard deviation of the mean

Results

Descriptive results

The descriptive statistics of the respondents are shown in Table 1. After deleting 123 questionnaires due to the responses to the categories not used for analysis, the final data used for analysis included 933 respondents. The average age of the respondents was about 43 years old and about half (51%) of them were male. About one-quarter (24%) of the respondents reported a monthly household income of 6 million KRW ($5,378) and above and about two-thirds (67%) of them were married. These socio-demographic statistics are very similar to those from the national census data.

Estimation results

The ordered probit model was estimated to examine public acceptance of offshore wind farms and the results are shown in Table 2. A goodness-of-fit index measured with McFadden's pseudo R^2 was 0.180. Among the explanatory variables included in the model, the following ones of gender, capital, visit, perceived benefits, perceived costs, human domination, economic impacts of tourism and social impacts of tourism were statistically significant at least at the 0.10 level. The positive coefficients of gender

Table 2. Results of ordered probit model – public acceptance as dependent variable.

Variable	Coefficient	Standard error	t-value
Perceived benefits	1.1361** [a]	0.087	13.09
Perceived costs	−0.1885**	0.053	−3.53
Balance of nature	−0.0499	0.086	−0.58
Human domination	0.0770	0.071	1.09
Economic impacts of tourism	−0.1156*	0.069	−1.68
Social impacts of tourism	0.1325*	0.077	1.72
Age	−0.0009	0.004	−0.21
Gender	0.3697**	0.078	4.75
Marriage	0.0969	0.109	0.89
Income	−0.0068	0.091	−0.07
Capital	0.1590**	0.077	2.06
Visit	0.1419*	0.078	1.82
Bill	0.0944	0.083	1.13
Log-Likelihood	−838.25	N = 933	
Pseudo R^2	0.155		

[a] *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

Table 3. Results of logit model – trip intention as dependent variable.

Variable	Coefficient	Standard error	t-value
Intercept	1.7194* [a]	0.945	1.82
Perceived benefits	−0.5642**	0.164	−3.45
Perceived costs	0.4441**	0.108	4.10
Balance of nature	0.4899**	0.166	2.95
Human domination	−0.0638	0.138	−0.46
Impacts on tourism	−0.2668*	0.140	−1.91
Impacts on seascape	−0.6630**	0.118	−5.62
Age	−0.0188**	0.008	−2.33
Gender	−0.3995**	0.147	−2.71
Married	0.2096	0.208	1.01
Income	0.0611	0.172	0.36
Visit	−0.1275	0.149	−0.86
Log-Likelihood	−559.65	N = 933	
Pseudo R^2	0.130		

[a] *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

(β = −.370), capital (β = .159) and visit (β = −.142) indicate that males, those who live in the capital region and those who have previously visited wind farms are more supportive of offshore wind farm projects than their counterparts. Likewise, the positive coefficients (β = −1.136) of perceived benefits and social impacts of tourism indicate that those who consider wind power positive as clean energy and those who think that tourism produces diverse social benefits are more favorable to offshore wind farm projects. The negative coefficients of perceived costs (β = −.189) and economic impacts of tourism (β = −.116), conversely, signify that those who perceive that wind power negatively affects the environment and those who think that tourism produces diverse economic benefits are less supportive of offshore wind farm projects.

Next, the logit model was estimated to explain which explanatory variables influence individuals' contingent trip behavior coded as 1 = decrease the number of trips to the beach if wind farms are installed and 0 = would not change or increase the number of trips (Table 3). McFadden's pseudo R^2 was 0.130. A total of six explanatory variables, age, gender, perceived benefits, perceived costs, balance of nature and impacts on seascape, were statistically significant at the 0.05 level and one variable of impacts on

Table 4. Results of multinomial logit model - site substitution as dependent variable.

	(Base = no site substitution[b])	
Variable	Nearby site substitutability	Distant site substitutability
Intercept	−0.6938 (1.198)	1.4960 (1.525)
Perceived benefits	−0.6649 (0.206)**[a]	−1.1947 (0.244)**
Perceived costs	0.7035 (0.143)**	0.9206 (0.182)**
Balance in nature	0.3077 (0.211)	−0.0579 (0.253)
Human domination	0.4632 (0.175)**	0.3772 (0.225)*
Impacts on tourism	−0.4208 (0.169)**	−0.5063 (0.202)**
Impacts on seascape	−0.8334 (0.154)**	−0.9194 (0.195)**
Age	−0.0014 (0.010)	0.0240 (0.013)*
Gender	−0.2455 (0.191)	−0.5749 (0.236)**
Married	0.3039 (0.270)	−0.4624 (0.335)
Income	−0.0894 (0.230)	0.3386 (0.263)
Visit	−0.3945 (0.192)**	−0.04293 (0.240)*
Log likelihood = −635.33		N = 933
Pseudo R^2		0.171

Note: Standard errors in parenthesis.
[a] $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.
[b] no site -substitution used as the reference level. The outcomes of the other two categories (i.e., nearby site substitution and distant site substitution are interpreted as a relative probability based on the reference level).

local tourism was significant at the 0.10 level. The positive coefficients of perceived costs and balance of nature indicate that individuals who perceive that wind power negatively affects the environment and those with a biospheric attitude are more likely to reduce their trips to the beach after the installation of wind farms. Conversely, the negative coefficients of age, gender, perceived benefits and impacts on local tourism and seascape denote that females, younger individuals, those who consider wind power positive as clean energy, and those who think that offshore wind farms positively affect local tourism and seascape are less likely to reduce their trips to the beach.

Finally, the multinomial logit model was estimated to examine which explanatory variables affect individuals' site substitution decision. This model included two alternative choice outcomes and one base level, helping researchers understand how the effects of the explanatory variables differ between individuals who are willing to substitute nearby destinations (in the same city or county) over no site substitution and those who are willing to substitute distant destinations (in a different city or county) over no site substitution (Table 4). McFadden's pseudo R^2 was 0.171.

The results show that the six explanatory variables, visit (β = −.394), perceived benefits (β = −.665), perceived costs (β = .704), human domination (β = .463), impacts on tourism (β = −.421) and impacts on seascape (β= −.833) were statistically significant at least at the 0.10 level in predicting individuals' willingness to substitute a beach site in the same city or county over no willingness to substitute. Thus, individuals who perceive that wind power negatively affects the environment and those with an anthropocentric attitude are more likely to substitute their original beach site with another in the same city or county. Conversely, individuals who have previously visited wind farms, those who perceive wind power positively as clean energy and those who think that offshore wind farms positively affect local tourism and seascape are less likely to substitute their original beach site with another in the same city or county.

Also, besides the variables of age (β = .024) and gender (β = −.575), the six variables, visit (β = −.429), perceived benefits (β = −1.195), perceived costs (β = .921),

human domination (β = .377), impacts on tourism (β = −.506) and impacts on seascape (β = −.919), were statistically significant at least at the 0.10 level in predicting the willingness to substitute the original beach site with another in a different city or county over no willingness to substitute. The same aforementioned interpretation can be applied to the six variables but females and older individuals are also more likely to substitute the site with another in a different city or county.

Discussion and conclusions

Given the mounting importance of wind energy for the transition to low-carbon societies, this article intended to answer the following research questions: (1) how important are various explanatory variables in shaping the public acceptance of offshore wind energy?, (2) how do these variables affect individuals' travel intention to visit the beaches when offshore wind farms are constructed nearby?, and (3) what are the roles of the explanatory variables in individuals' site substitution upon their forthcoming beach trips? Study results show that individuals' perceptions of wind energy and attitudes toward tourism were the critical variables that help understand the public acceptance of wind energy. Likewise, individuals' trip intention was closely connected with their perceptions of wind energy, attitudes toward tourism and biospheric attitude. Finally, the variables of individuals' perceptions of wind energy and attitudes toward tourism as well as their anthropocentric attitude played an important role in explaining their site substitution decision. Based on the results, several key points that can provide policy-related implications are worth noting below.

First, environmental attitudes play an important role in affecting their trip behaviors. Thus, individuals with a more environmentally friendly view (i.e., a biospheric attitude) are likely to reduce their trips to beaches nearby with wind farms and those with an anthropocentric attitude are likely to choose the beach destination without offshore wind farms nearby. Nevertheless, these variables were not important in explaining the public acceptance. These results may appear perplexing but can be understood in the context that individuals' decisions rely on at least two different preference orderings: private preferences that seek their maximized utility and public preferences that reflect more values about what is best for society (Ek 2005; Sagoff 1988). Rand and Hoen (2017) assert that environmental attitudes can be correlated with either the conservation of the natural landscape and wildlife or the benefits of renewable energy depending on the environmental impacts of offshore wind power and energy policy priorities. As a result, the effects of environmental attitudes on the public acceptance have not been clearly determined in previous studies. Environmental attitudes affect individuals' tourism behaviors differently as they make decisions relying on private preference. Individuals with strong environmental attitudes probably think they can control their own trip decisions. In particular, those with a strong biospheric attitude are likely to act more receptively by reducing the number of trips to the beach with offshore wind farms nearby but those with a strongly anthropocentric attitude more actively choose another beach destination without the sight of offshore wind farms in order to maximize their utility. Furthermore, given differences in general and specific attitude objects, the concepts of the social gap and the individual gap may help explain the results (Bell, Gray, and Haggett 2005; Bell et al. 2013). While the social gap is closely

connected with a biospheric perspective of the environment, those with a human centered view tend to oppose the installation of offshore wind farms more actively by not visiting their original destination.

Second, individuals' perceptions of wind energy (i.e., perceived benefits and costs) are the single most important explanatory variable across the three dependent variables. Study results are compatible with prior research (e.g., Leiren et al. 2020) and indicate that individuals who perceive wind energy positively as clean energy are likely to support wind farm projects more and are less likely to change their visits to the beaches in terms of trip frequency and destination even after the development of offshore wind farms nearby. Likewise, individuals who think that wind farms harmfully affect wildlife and nature conservation as well as human health and wellbeing are less positive about offshore wind power and tend to change their trip patterns more actively.

While these two variables are understood as individuals' general perceptions of societal impacts, people generally value the benefits of wind power more highly than the costs based on the mean values of perceived benefits and costs (3.76 and 3.16, respectively). This suggests that people largely support offshore wind power projects but are also concerned about the perceived costs of societal impacts. Multiple offshore wind farm projects are currently planned to be built in every major coastal city of South Korea and many other countries; however, many of them have been delayed (or canceled in some extreme cases) due to a lack of public acceptance. It is challenging to change individuals' perceptions of wind energy quickly but the results of the acceptance model show that two variables could play an important role in shaping perceptions: their experience of visiting wind farms and residence in the capital region. Individuals' personal experience with wind turbines may not necessarily be positive and facilitate public acceptance. However, it is known that negative perceptions often result from a lack of knowledge and experience and people feel that the impacts of wind power are less harmful than initially thought as familiarity with wind power increases (Ladenburg 2010; Wolsink 2018). Also, the residents of the capital region have generally been more exposed to environmental campaigns and demonstrations (Kim, Nam, and Yoo 2020). Consequently, improving the access of those living outside the capital region to information and understanding wind power is therefore required so that these people can also experience the benefits of wind power directly or indirectly, increasing the familiarity of these individuals with wind farms.

Third, although some individuals may not care about or prefer the presence of wind farms, their evaluations of offshore wind farms on local tourism and seascape significantly affect their trip behavior and site substitution. Landscape and esthetic value and tourism and recreation value consist of a principal part of cultural services provided by the beaches. Thus, concerns about the visual and audial impacts of wind power are closely linked with annoyance and objections to wind projects particularly from the perspective of tourism and recreational activities (Rand and Hoen 2017; Westerlud 2020). It is acknowledged that individuals tend to express more concern about the presence of wind turbines at tourism and recreational sites including beaches than those in residential areas (Leiren et al. 2020). In South Korea, most offshore wind farms have been built at relatively short distances, about 1 km or less from the coast (Park and Kim 2019), rendering the wind turbines easily visible from the beach. As a result, the presence of wind turbines at this distance are likely to negatively affect

coastal visitors by making them visit the original beach less and/or forcing them to substitute their original beach site with another one with less or no visual impact. The changing visitation patterns of coastal tourists likely cause negative economic impacts derived from foregone revenues and in the destination and thus decision-makers and wind energy professionals need to take account of the responses of diverse stakeholders including coastal tourists on siting offshore wind farms. Feasible alternatives may include avoiding popular beaches that are located in major coastal destinations or increasing the distance from the coast by developing floating wind turbines that have recently attracted attention.

Based on the Green New Deal policy that seeks low carbon and eco-friendly economic growth through renewable energy and its related industrial infrastructure and the RE100 plan, which aims to replace 100% of the electricity used by enterprises with renewable energy by 2050 (RE100, 2021), offshore wind power has been drawing more attention than ever. Previous research has focused mainly on the public acceptance of offshore wind power particularly from the perspective of residents who reside on or close to the coast. This study, however, suggests that besides coastal residents, other local residents and tourists, who have used beaches as primarily for coastal recreation and tourism need to be included as stakeholders in the decision-making process of the installations of offshore wind power. While this study points out that cultural ecosystem services need to be considered as one of primary components in the decision-making process, diverse future research will be also beneficial on the social, economic and environmental benefits and costs derived from offshore wind farms.

In Korea, the government recently announced an increase to the renewable energy credits on offshore wind farms in order to help to place wind turbines in more deep-sea areas. While this policy change seems to be encouraging, numerous offshore wind farm projects, which are located closed to the coast, are still in progress. The results of this study are expected to provide some meaningful implications that can help minimize the potential conflicts among stakeholders and prevent any loss in economic impacts due to individuals' modified visit patterns due to the siting of offshore wind farms.

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